

George Holland

December 1985

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In 1948 K. W. Neatby, Director of Science Service, invited Dr. Holland to become Chief of the Systematic Entomology Unit of the Entomology Division and Curator of the Canadian National Collection of Insects in Ottawa, Ontario. In this position he was responsible for identifications and taxonomic research for the Department. In 1955 he was appointed Chief of Insect Systematics and Biological Control Unit of the Entomology Division. In 1959 he was made the Director of the Entomology Research Institute, Research Branch, a position he occupied until poor health compelled him to step down in 1969. However, he retained a research position in the Institute to continue his work on Siphonaptera. During his tenure of leadership, the entomology unit with six

scientists developed into an institute with 45, the Canadian National Collection grew from about one million to seven million specimens and the status and productivity of the E.R.I. and the scope of its collections grew to one of the top three in the world.

Other important offices occupied by Dr. Holland include:
Chairman, Program Committee, Tenth International Congress of Entomology, Montreal, 1956
Chairman, Defence Research Board Advisory Committee on Entomological Research, 1963-1967
Chairman, Centennial of Entomology in Canada, 1863 to 1963, 1963.

He was also the Canadian member of the Permanent Committee of the International Congresses of Entomology. Field work and attendance at international conferences involved him in extensive travel in the United States, the United Kingdom, Hawaiian and Figi Islands, Australia, Territory of Papua and New Guinea, the Soviet Union and Japan.

Dr. Holland was active in a number of professional and scientific societies. He was a Fellow of the Royal Entomological Society of London and the Royal Society of Canada. As a member of the Entomological Society of Canada he served as president in 1957-1958; was made an Honorary Member in 1973, a Fellow in 1976 and in 1979 was presented the E.S.C. Gold Medal for Outstanding Achievement in Canadian Entomology. He was a member of the Entomological Society of America and in the 1950's and 1960's served on committees on Entomological Nomenclature and International Relations and chaired the Taxonomy Section in 1952. He belonged to the Entomological Society of Ontario, was Director in 1949-1950 and elected a Fellow in 1971. He was also a member of the Entomological Society of British Columbia, Society of Systematic Zoology, Canadian Society of Zoologists and the Australian Mammal Society.

In spite of heavy administrative responsibilities, Dr. Holland was still able to publish over 50 technical studies including two books on the fleas of Canada in 1949 and 1985, and a work on the fleas of New Guinea in 1969. A copy of his bibliography appears at the end of this issue.

Dr. Holland is survived by his wife, a son, two daughters and four grandchildren.

Those of us who counted George Holland as a friend as well as a colleague remember him as a witty, generous person who always could be counted on for assistance. His passage is a great loss to the scientific community as well as to his friends.

During his career of studying the Siphonaptera, Dr. Holland jointly or solely described 8 genera and 70 species and subspecies. These are listed below, along with their current status.

Genera

Aetheopsylla Stewart & Holland, 1940 = Oropsylla Wagner & Ioff, 1926
Orthopsylloides Holland, 1969
Parastivalius Holland, 1969
Papuapsylla Holland, 1969

Striopsylla Holland, 1969
Metastivalius Holland, 1969
Obtusifrontia Holland, 1969
Rectidigitus Holland, 1969

Species

septentrionalis Stewart & Holland, 1940 Aetheopsylla = Oropsylla arctomys
(Baker, 1904)
tundrensis Holland, 1945 Ceratophyllus
occidentalis Holland, 1949 Hystrichopsylla
spinata Holland, 1949 Hystrichopsylla
campestris Holland, 1949 Callistopsyllus
obscurus Holland, 1949 Megabothris = Megabothris clantoni Hubbard, 1949
pacifica Holland, 1949 Peromyscopsylla hesperomys
hamata Holland & Jameson, 1950 Nearctopsylla
princei Holland & Jameson, 1950 Nearctopsylla
gregsoni Holland, 1950 Megabothris calcarifer = Megabothris asio gregsoni
fulleri Holland, 1951 Catallagia dacenkoi = Catallagia dacenkoi Ioff, 1940
lari Holland, 1951 Ceratophyllus
scopulorum Holland, 1952 Ceratophyllus
arcuegens Holland, 1952 Ceratophyllus
athabasca Holland, 1952 Malaraeus penicilliger = Amalaraeus dissimilis athabasca
traubi Holland & Jellison, 1952 Mioctenopsylla
jellisoni Holland, 1954 Catallagia
neotomae Holland, 1957 Hystrichopsylla dippiei
truncata Holland, 1957 Hystrichopsylla dippiei
linsdalei Holland, 1957 Hystrichopsylla
neweyi Holland & Loshbaugh, 1958 Catallagia
nearctica Holland & Loshbaugh, 1958 Ornithophaga = Ornithophaga anomala Mikulin,
1957
arizonae Holland, 1960 Catallagia
rauschi Holland, 1960 Ceratophyllus
martini Holland, 1965 Anomiopsyllus
mexicanus Holland, 1965 Anomiopsyllus hiemalis
durangoensis Holland, 1965 Anomiopsyllus
sinuatus Holland, 1965 Anomiopsyllus
papuanus Holland, 1969 Porribius
aiyurensis Holland, 1969 Acanthopsylla
monticola Holland, 1969 Acanthopsylla
eudromiciae Holland, 1969 Acanthopsylla
bisinuata Holland, 1969 Acanthopsylla
novaeguineae Holland, 1969 Hoogstraalia
spinata Holland, 1969 Pygiopsylla
bismarckensis Holland, 1969 Pygiopsylla laciniosa
archboldi Holland, 1969 Pygiopsylla
smittiana Holland, 1969 Pygiopsylla
traubi Holland, 1969 Pygiopsylla
vandeuseni Holland, 1969 Traubia
peristigma Holland, 1969 Traubia
kuru Holland, 1969 Traubia
orthodactylus Holland, 1969 Orthopsylloides orthodactylus
owiensis Holland, 1969 Orthopsylloides orthodactylus
brassi Holland, 1969 Parastivalius
gressitti Holland, 1969 Parastivalius

luluae Holland, 1969 Papuapsylla
barretti Holland, 1969 Papuapsylla
minor Holland, 1969 Papuapsylla
vandeuseni Holland, 1969 Striopsylla
huonensis Holland, 1969 Metastivalius
rothschildae Holland, 1969 Metastivalius
tamberan Holland, 1969 Metastivalius
simplex Holland, 1969 Obtusifrontia
traubi Holland, 1969 Rectidigitus
szentivanyi Holland, 1969 Rectidigitus
globata Holland, 1971 Myodopsylla
martini Holland, 1971 Kohlsia
brevipecten Holland, 1971 Austropsylla
multidentata Holland, 1971 Acanthopsylla rothschildi
dunneti Holland, 1971 Acanthopsylla
incerta Holland, 1971 Acanthopsylla = Acanthopsylla pavida (Roths., 1916)
rectangulata Holland, 1971 Acanthopsylla scintilla
tasmanica Holland, 1971 Acanthopsylla scintilla
arcuata Holland, 1971 Pygiopsylla
sinuata Holland, 1971 Pygiopsylla
iridis Holland, 1971 Pygiopsylla
grahami Holland, 1979 Nearctopsylla
rauschi Holland, 1979 Rhadinopsylla (Micropsylla)
calderwoodi Holland, 1979 Ceratophyllus (Ceratophyllus)

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CONGRATULATIONS - S. G. MEDVEDEV successfully defended his doctoral dissertation in March 1985 at the Zoological Institute of the Academy of Sciences USSR (Leningrad). His dissertation is entitled "The fleas fam. Ischnopsyllidae (Siphonaptera) USSR". Portions of his work were published in 1984 in Entomol. Obozr. English translations of these papers should be available soon in Entomol. Review.

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VISITORS - We have had the following visitors since Flea News 30 was mailed:

Dr. LI K.-C. and her husband, Dr. CHIN T.-H., of Guiyang Medical College, Guiyang, People's Republic of China, spent the weekend of June 15-16 working in the collection.

Dr. JENELLA E. LOYE stopped by in mid-August on her return trip to the University of Utah, Salt Lake City, Utah.

Dr. ROBERT TRAUB spent October 5-9 in Ames. During his visit he presented seminars in both Entomology and Veterinary Medicine. His visit was jointly supported by the Department of Entomology, the College of Veterinary Medicine and the Graduate College.

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Answer to FLEA TRIVIA QUESTION in Flea News 30: 2 (Pulex irritans and Tunga penetrans)

RECENTLY PUBLISHED - Synopsis and Classification of Living Organisms. Sybil P. Parker (Editor in Chief), 1982. (ISBN 0-07-079031-0 (set)). McGraw-Hill Book Company (New York, etc.). Vol. 1, i-xviii, 1-1166; Vol. 2, 1-1232. US \$175.00 (2 volume set).

As the title proclaims, these monumental volumes present a classification of all known living organisms, from viruses to vertebrates. The bulk of the volumes consists of 8200 synoptic articles which give a brief overview of the characteristics, biology, ecology and distribution of the various taxa. The classification extends from the two superkingdoms, the Prokaryotae and Eukaryotae, down to the family level. Most family treatments include the number of known genera and the major genera may be named and briefly discussed. Distributional data are also included. Each synopsis ends with one or more literature citations which may be consulted for additional information.

It is a recognized paradox that there are almost as many biological classifications as there are classifiers. Modern biologists arrange the living world in different ways, from the plant and animal kingdoms in the simplest classification to a system where the prokaryotes, protists and fungi are relegated to their own separate kingdoms. The system followed here recognizes only 4 kingdoms: Virus, Monera, Plantae and Animalia. Some might argue that the Protista should be recognized as a separate kingdom but the system followed in this work seems quite satisfactory. An examination of the two groups of organisms with which I am most familiar, namely, insects and vertebrates, revealed a relatively conservative approach to the subject. Of particular interest to readers of this newsletter is the classification of the Siphonaptera by F.G.A.M. Smit which is discussed in more detail in this issue.

This is not the type of book that most of us would purchase for our personal libraries, particularly at that price, but it is comforting for me to know that such a compendium is as close as the reference room of our library.

---R.E.L.

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RECENTLY PUBLISHED - International Code of Zoological Nomenclature, Third Edition, adopted by the XX General Assembly of the International Union of Biological Sciences, 1985. i-xx, 338 pp. Available from the International Trust for Zoological Nomenclature in association with the British Museum (Natural History), London, and University of California Press, Berkeley and Los Angeles.

A thorough review of this new edition of the Code is not practical here, but it is a vast improvement over the 1963 second edition. While it is arranged in the same order as the earlier edition, each section has been expanded and additional subheadings and examples have been included.

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A REVISED CLASSIFICATION OF SIPHONAPTERA - In his personal bibliography which accompanied Flea News 20, Frans Smit included the following citation:

1980f - Siphonaptera. In: Taxonomy and classification of living organisms. New York (McGraw-Hill Co.). [in press]

In the ensuing years I have often thought of writing to Frans or McGraw-Hill to find out the fate of that particular publication. Since there seemed to be no urgency to do this and since there are only a few things upon which I can procrastinate, nothing was ever done. Then, recently, while doing her thing in the library, J.H.L. happened onto a 2-volume set with a suspiciously familiar sounding title: "Synopsis and Classification of Living Organisms". (A brief review of these volumes appears elsewhere in this issue of Flea News.) A cursory examination revealed that the section on the Siphonaptera had indeed been written by F.G.A.M. Smit and that it contained a classification of the order which differed somewhat from that in use during the past few decades. A diagrammatic representation of the Smit classification is presented here along with a brief discussion.

First, the order is divided into 5 superfamilies which collectively contain 15 families. The families Coptosyllidae, Ancistropsyllidae, Xiphiopsyllidae and Malacopsyllidae remain as originally conceived and have no subfamilies. The Tungidae has been resurrected as a full family, also without subfamilies, and contains only the genus Tunga. The Stephanocircidae, Chim-aeropsyllidae, Ischnopsyllidae and Rhopalopsyllidae are also unaltered and still contain their traditional subfamilies. The remaining families have been altered in their subfamily content.

The Hystrichopsyllidae has been reduced to 2 subfamilies. The Hystrichopsyllinae contains the 4 hystrichopsyllid genera the females of which have paired spermathecae (Hystrichopsylla, Atyphloceras, Typhloceras and Ctenoparia). The Macropsyllidae has been reduced to subfamily status and constitutes the other subfamily. Females of both Macropsylla and Stephanopsylla also have 2 spermathecae. All of the remaining subfamilies normally assigned to the Hystrichopsyllidae are grouped under a new family name, the Ctenophthalmidae.

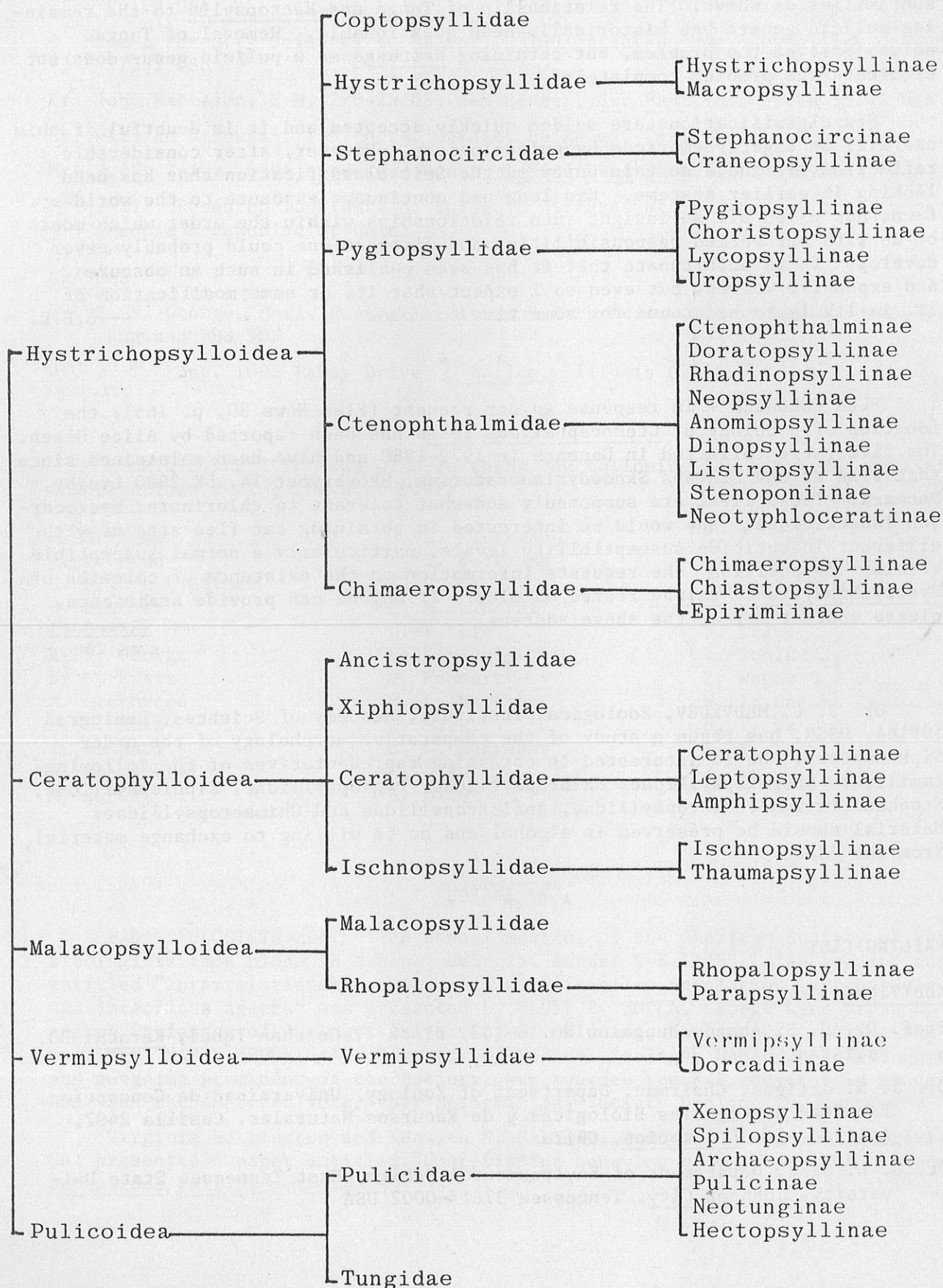
The Pygiopsyllidae contains 4 subfamilies. The Lycopsyllinae and Pygiopsyllinae were previously recognized and the tribes Choristopsyllini and Uropsyllini have been elevated to subfamily rank. The previously recognized Stivaliinae has been eliminated.

As pointed out by Smit (1983), the Ceratophyllidae and the Leptopsyllidae are closely related and there is some overlap in certain characters. As a result, in this classification the Leptopsyllidae is eliminated and the subfamilies Leptopsyllinae and Amphipsyllinae are placed in the Ceratophyllidae along with the Ceratophyllinae. (Those noting the discrepancy between Smit's 1982 classification and his 1983 treatment of the Ceratophyllidae in Traub, Rothschild and Haddow should be aware that the latter was written in the mid 1970's but not published until 1983. In the intervening years there was ample time for Smit to further firm up his ideas on subordinal classification.)

The Vermipsyllidae has been divided into 2 subfamilies: the Vermipsyllinae for Vermipsylla and Chaetopsylla and the Dorcadiinae for Dorcadia.

In the Pulicidae, removal of Tunga to its own family allowed the elevation of the remaining 5 tribes to subfamily status. Retaining Neotunga in the

CLASSIFICATION OF THE SIPHONAPTERA
 (after Smit, 1982. In: Parker, S. P. (Ed.)
 Synopsis and classification of living organisms.
 Vol. 2. McGraw-Hill, New York, pp. 557-563.)



Pulicidae necessitated the erection of a subfamily for it, resulting in the 6 subfamilies as shown. The relationship of Tunga and Hectopsylla to the remaining pulicid genera has historically been questionable. Removal of Tunga solves part of the problem, but retaining Neotunga as a pulicid genus does not eliminate the problem completely.

New classifications are seldom quickly accepted and it is doubtful if this one will be eagerly embraced by pulicologists. However, after considerable reflection, I find a certain unity in the Smit classification that has been lacking in earlier systems. His long and continuous exposure to the world fauna has given him an insight into relationships within the order which most of us with our varied responsibilities and distractions could probably never develop. It is unfortunate that it has been published in such an obscure and expensive volume, but even so I expect that it, or some modification of it, is likely to be around for some time to come. ---R.E.L.

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FLEA COLONIES - In response to our request (Flea News 30, p. 165), the location of a colony of Ctenocephalides felis has been reported by Alice Olsen. The fleas were collected in Denmark in 1979-1980 and have been maintained since that time at the Statens Skadedyrlaboratorium, Skovbrynet 14, DK 2800 Lyngby, Denmark. These fleas are supposedly somewhat tolerant to chlorinated hydrocarbon insecticides. She would be interested in obtaining cat flea strains with different insecticide susceptibility levels, particularly a normal susceptible strain. In addition, she requests information on the existence of colonies of Hystrihopsylla t. talpae reared on mice. If anyone can provide assistance, please contact her at the above address.

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Dr. S. G. MEDVEDEV, Zoological Institute, Academy of Sciences, Leningrad 199164, USSR, has begun a study of the comparative morphology of the order Siphonaptera. He is interested in obtaining representatives of the following families: Rhopalopsyllidae, Malacopsyllidae, Pygiopsyllidae, Xiphiopsyllidae, Stephanocircidae, Macropsyllidae, Ancistropsyllidae and Chimaeropsyllidae. Material should be preserved in alcohol and he is willing to exchange material from the USSR.

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MAILING LIST

ADDITIONS

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Dr. J. T. Lang, 1002 Nancy Drive, O'Fallon, Illinois 62269 USA

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REPRINTS RECEIVED - We wish to thank the following for supplying reprints and citations during the past year:

J. Artigas	G. E. Kietzmann	E. K. Saliba
A. F. Azad	W. B. Kottkamp	V. Sapegina
G. Brinck-Lindroth	K.-C. Li	T. G. Schwan
B. Drucker	P. M. Linardi	J. Segerman
A. Dudich	C.-Y. Liu	R. Traub
R. S. George	L. Lundqvist	K. Uchikawa
N. G. Gratz	V. Mahnert	Z. Wegner
M. Greenwood	H. L. Martin	World Health Organization
G. E. Haas	C. Maser	H.-Y. Wu
T. R. Hadi	S. G. Medvedev	R.-Y. Ye
H. Hoogstraal	W. Olkowski	X. Yu
M. Jurik	A. Olsen	J.-T. Zhang

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CONFERENCES

PARASITOLOGISTS MEET - The annual meeting of the American Society of Parasitologists took place in Athens, Georgia, August 4-8, 1985. The opening address entitled "Interrelationships between a cliff swallow population, ectoparasites, and infectious agents" was presented by CLUFF E. HOPLA, George Lynn Cross Research Professor of Zoology, University of Oklahoma, Norman, OK. During the meetings, HARRY HOOGSTRAAL, Director of Medical Zoology, NAMRU 3, Cairo, Egypt, and outgoing president of the Society, was awarded the K.F. Meyer Gold Headed Cane and the American Veterinary Epidemiology Society Diploma.

Virginia Herrington and Kenneth S. Saladin, Georgia College, Milledgeville, GA, presented a paper entitled "Host-finding behavior of the cat flea, Ctenocephalides felis".

SVE - The Society of Vector Ecologists held their 17th Annual Conference, November 20-22, 1985, at the University of California, Irvine. Among the many interesting papers presented, the following are of particular interest to pulicologists.

Symposium: Forensic Entomology. Moderator: M. L. Goff.

Dermatologic manifestations of ectoparasites and their forensic implications.
J. Pritchard, M.D., Ventura County General Hospital, California.

Colloquium: Flies, fleas, bugs & todes. Moderator: G. C. Collett.

Transmission of plague, *Yersinia pestis*, to the northern grasshopper mouse, *Onychomys leucogaster*, by the flea *Thrassis fatus*.

R. E. Thomas, Department of Zoology, University of Oklahoma, Norman, OK
G. O. Maupin, Plague Branch, Division of Vector-borne Diseases, CDC,
Fort Collins, CO.

Murine typhus in Los Angeles County.

A. Tilzer, Vector Control Programs, Los Angeles County Department of
Health Services, CA.

Symposium: Ecology of ectoparasites. Moderator: C. E. HOPLA.

Fleas associated with ground squirrels in the western United States.

G. O. Maupin, Plague Branch, Division of Vector-borne Diseases, CDC,
Fort Collins, CO.

ESA - The Entomological Society of America National Conference will be held December 8-12, 1985, at Hollywood, Florida. The program lists two papers on fleas.

Ectoparasites: Biology and control.

J. A. DeVane, ARS-USDA, Veterinary Toxicology and Entomology Research
Laboratory, College Station, TX.

Fleas, IGR's and resistance.

P. G. Koehler, Department of Entomology & Nematology, University of
Florida, Gainesville, FL.

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PROVISIONAL LISTS OF TAXA DESCRIBED IN 1984 AND 1985

1984

Rhadinopsylla (*Actenophthalmus*) *biconcava* Chen, Ji & Wu
Nearctopsylla (*Neochinopsylla*) *liupanshanensis* Li, Wu & Liu
Lagaropsylla mytila Hürka
Parapsyllus magellanicus largificus Smit
Agastopsylla nylota euneomys Lewis
Hystrihopsylla weida qinlingensis Zhang, Wu & Liu
Palaeopsylla brevifrontata Zhang, Wu & Liu
Palaeopsylla anserocephoides Zhang, Wu & Liu
Geusibia minutiprominula minutiprominula Zhang & Liu
Geusibia minutiprominula ningshanensis Zhang & Liu
Geusibia longihilla Zhang & Liu
Callopsylla (*Callopsylla*) *digitata* Cai, Wu & Liu
Leptopsylla (*Pectinoctenus*) *pectiniceps ventrisinulata* Chen, Zhang & Liu
Ischnopsyllus (*Hexactenopsylla*) *jinciensis* Xiao

(1976 - 1981)

1985

Peromyscopsylla himalaica australishaanxia Zhang & Liu
Peromyscopsylla scaliforma Zhang & Liu
Paradoxopsyllus calceiforma Zhang & Liu
Liuopsylla conica Zhang, Wu & Liu
Sigmactenus celebensis Lewis & Jones
Gryphopsylla hetera Lewis & Jones
Stivalius exoticus Lewis & Jones
Typhloceras poppei orientalis Mahnert & Beaucournu
Rhadinopsylla (Actenophthalmus) stenofronta Xie
Frontopsylla rotunditruncata Cai & Liu
Typhlomyopsyllus esinus Liu, Shi & Liu
Vermipsylla parallela rhinopitheca Li

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FLEA TRIVIA QUESTION: In what year was the largest number of new taxa described and what was the number of names? Answer in Flea News 32!

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LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 18)

Kodama, J. K., J. A. Collins & K. R. van Kampen - Toxic reactions due to flea collars. J. amer. vet. Med. Assoc. 168: 368-374

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 17)

Doll, J. M., P. M. Hotchkiss & M. E. Wright - Bubonic plague in Arizona. Ariz. Morb. 4(10): 1-4

*Quintera, M. T. & A. Acevedo - Frecuencia e importancia de las pulgas - Ctenocephalides felis y Pulex irritans en granjas de gallinas. Mem. IV Convencion Anual Asoc. Nal. de Especialistas en Cienc. Avicolas A. C. Pto Vallarta, Jal., 2-5 Mayo 1979.

*Tracy, C. H. - Fleas and flea control. Today's Animal Health 8: 10-12

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 13)

*Anonymous - The plague, "black death". Med. Bull. U.S. Army Europe 37(7): 14-17

*Kahle, J. - Fleas and how to control them. NCAP News, Winter-Spring: 46-51

*Lorenz, M. D. - The management of flea allergy dermatitis. In: Kirk, R. W., ed., Current Veterinary Therapy VII. W. B. Saunders Co., Philadelphia, pp. 446-449

Rail, C. D., T. L. Hanna & W. R. Moore - Plague, a growing concern. J. environ. Health 42(6): 315-319

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 11)

*Fadok, V. A. - Ectoparasites of the dog and cat. AHT [Animal Health Technician?] 2(6): 292-300

(1981 - 1983)

- *Gadzhiev, A. T., T. A. Dubovchenko, Z. A. Mustafaeva & M. G. Kuliev - Ectoparasites of synanthropic animals (rodents, bats, birds) and cases of their infestation of man in Azerbaijan. In: Tagi-zade, T. A., ed., Aktual'nye voprosy meditsinskoj parazitologii i tropicheskoi meditsiny. Chast' I: 66-68
- *Golovach, G. P. - Quantitative characteristics of fleas (Siphonaptera) in winter nests of the European bank vole (*Clethrionomys glareolus*) in the territory of the Kanev Reserve. In: Topchii, G. L., Ekologo-morfologicheskie osobennosti zhivotnykh i sreda ikh obitaniya, 'Naukova Dumka', Kiev
- *Kilonzo, B. S. - The origin, dissemination and present status of plague in Tanzania. Dar Es Salaam med. J. 8(2): 130-142
- *Komba, E. - Plague: a growing public health problem in Tanzania? Dar Es Salaam med. J. 8(2): 59
- Olsen, P. F. - 19 Sylvatic Plague. In: Davis, J. W., L. H. Karstad & D. O. Trainer, Infectious Diseases of Wild Mammals, 2nd ed., Iowa State Univ. Press, Ames, pp. 232-243

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1982 (list 8)

- *Ehmann, N. & K. O. Story - Chapter 18. In: Mallis, A., Handbook of Pest Control, 6th Ed., Franzak & Foster Co., Cleveland, Ohio, pp. 603-618
- Hiepe, P. T. - Large scale management systems and parasite populations: ectoparasites. Vet. Parasitol. 11: 61-68 [Siph.: 66]
- Olsen, A. & N. D. Christensen - Fleas. Danish Pest Infestation Laboratory Annual Report, pp. 74-83
- *Rifaat, M. A., T. A. Morsy & M. M. A. Mawla - Ectoparasites of rodents in Ismailiya Governate, Egypt. J. egypt. Soc. Parasitol. 12(1): 1-8
- Smit, F. G. A. M. - Siphonaptera. In: Parker, S. P., ed., Synopsis and classification of living organisms. Vol. 2. McGraw-Hill, New York, pp. 557-563
- Takahashi, M., S. Yamamoto, T. Saito & K. Machida - The fleas (Siphonaptera) parasitic on mammals and birds in Saitama and Gunma Prefectures, Central Japan. I. The fleas found on median and large sized mammals. Ann. Rep. Ohara General Hospital 25(32): 7-24
- WHO Expert Committee with FAO - Bacterial and viral zoonoses. WHO Tech. Rep. Ser. 682: 1-146

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1983 (list 6)

- Georghiou, G. P. & T. Saito - Pest resistance to pesticides. Plenum Press, New York, 809 pp. [Siph.: 45-46]
- Haitlinger, R. - Arthropod communities. In: Petrusewicz, K., ed., Ecology of the bank vole. Acta Theriol. 28, Suppl. 1: 55-68
- Intravartolo, C. - Client education. Fleas and flea control. Anim. Hlth. Tech. [title subsequently changed to Veterinary Technician]4(5): 281-282

(1983 - 1984)

- Janion, S. M. - Dynamics of an ectoparasite-host system. In: Petruszewicz, K., ed., Ecology of the bank vole. Acta Theriol. 28, Suppl. 1: 69-73
- *Khalil, M. S., A. M. El-Ridi & M. S. Arafa - The influence of host age on the survival and frequency of feeding of adult *Xenopsylla cheopis*. J. Egypt. Soc. Parasitol. 13(2): 481-484
- *Medleau, L. & W. H. Miller, Jr. - Flea infestation and its control. Internatl. J. Dermatol. 22(6): 378-379
- Munger, J. C., M. A. Bowers & W. T. Jones - Desert rodent populations: factors affecting abundance, distribution and genetic structure. In: Biology of desert rodents. Great Basin Nat. Mem. 7: 91-116
- Olkowski, W., H. Olkowski & S. Daar - IPM for the cat flea. IPM Practitioner 5(9): 7-11
- Olsen, A. - Fleas. Danish Pest Infestation Laboratory Ann. Rep., pp. 57-63
- Starikov, V. P. & V. F. Sapegina - Peculiarities in the distribution of fleas of small mammals in the Transural and Pre-Ural forest-steppe region. In: Dynamics of the numbers and role of insects in biogeocenoses of the Urals. Sverdlovsk, Akad. Nauk. SSSR, Ural. Nauch Centre: 47-48

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1984 (list 4)

- *Alexander, J. O'D. - Arthropods and human skin. Springer-Verlag, Berlin, 422 pp.
- Anderson, S. & J. K. Jones (Eds.) - Orders and families of recent mammals of the world. John Wiley & Sons, New York (sponsored by Amer. Soc. Mammal.), vii-xii, 686 pp.
- *Anonymous - Plague off to a fast start in 1984. Calif. Morb. May 25, 1984
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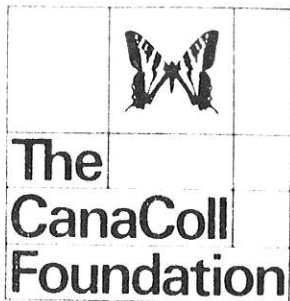
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

Flea News sends our Best Wishes for the New Year



To promote taxonomic research on the Canadian National Collection of insects and related arthropods

The George P. Holland Memorial Fund

Most of you know the tragic news that George P. Holland, 14 Kitimat Crescent, Nepean, Ontario K2H 7G5, died on 10 November 1985 from exposure during a hunting trip to his favorite lodge near Ottawa. George retired in 1976 and since has devoted much of his time to finishing his monograph on the fleas of Canada, Alaska, and Greenland (published in early 1985).

George was known world-wide for his many contributions to the taxonomy of fleas. He began his career in entomology at the Canada Department of Agriculture laboratory at Kamloops, British Columbia, and then moved to Ottawa to serve as head of the old Systematic Entomology Unit and then as director of the Entomology (now Bio-systematics) Research Institute.

Under George's leadership, the Canadian National Collection (CNC) of insects and related arthropods grew to be one of the best collections of its kind in the world with extensive nearctic holdings, especially from the arctic region, and with good collections from other areas of the world, such as New Guinea, Mexico, Jamaica, Europe, etc. During his tenure as director, the CNC grew from a relatively small collection of about 1,000,000 specimens to one with about 10,000,000 specimens.

The George P. Holland Memorial Fund has been established by the CanaColl Foundation in his memory. The fund is ongoing and can be added to at any time. CanaColl is an independent organization devoted to promoting research and development of the CNC. The accrued interest from this memorial fund will be used to support curation on the CNC.

Donations in memory of George may be sent to the CanaColl Foundation (address below); receipts for income tax purposes will be sent for all donations of \$5.00 or more. The Holland family will be informed of the names of those who have made donations to the fund.

Please share this notice with anyone else who might be interested. If you want more information, please don't hesitate to ask.

May this and other sentiments from friends be a source of comfort to the Holland family at this time.

15 November 1985

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Information concerning Dr. Holland was drawn from various sources, but mainly from biographic information provided by him in November 1980.

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